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Summary

Integrated M.S./Ph.D. researcher at the Korea University NLP Lab (Advisor: Prof. Heuseok Lim), specializing in large language models and natural language processing — with a focus on agentic LLMs, efficient reasoning, machine translation, information retrieval, and model merging. First-author research has been published at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**. Four papers were accepted at **EMNLP 2026** (2 Main, 2 Findings), with co-authored work at **ICLR 2026** and a series of preprints under review. Recipient of the **Best Paper Award (Top 1)** at HCLT 2025. Served as the post-training Project Manager and Korea University representative for **VAETKI**, Korea’s national Sovereign AI Foundation Model. Author of 30+ peer-reviewed papers across top-tier NLP venues and journals.

Education

Integrated M.S./Ph.D. in Computer Science

Korea University, Seoul, Korea

2021 – Present

Expected: Feb. 2027

- NLP Lab, Advisor: Prof. Heuseok Lim.
- Research focus: large language models, natural language processing, agentic LLM, efficient reasoning, machine translation, information retrieval, and model merging.

Research Interests

Large Language Models · Natural Language Processing · Agentic LLM · Efficient Reasoning · Machine Translation · Information Retrieval · Model Merging

Work Experience & Major Projects

VAETKI — Korean Sovereign AI Foundation Model

Post-training Project Manager — Korea University Representative

Jun 2025 – Jan 2026

NC AI (NCSOFT)-led consortium

- Served as post-training PM representing Korea University in Korea’s national **Sovereign AI Foundation Model** program, organized by the Ministry of Science and ICT — an approximately KRW 200 billion national initiative with staged six-month evaluations running through 2027.
- The Korea University team (Prof. Heuseok Lim) was selected as a core research partner in the NC AI (NCSOFT)-led grand consortium of 14 industry–academia–research institutions and 40 adopters (including POSCO DX, Lotte Innovate, KAIST, ETRI, and MBC).
- Led the full post-training pipeline — Supervised Fine-Tuning (SFT), Reinforcement Learning, and dataset curation — for the VAETKI model.
- **VAETKI model**: an industry-specialized sovereign foundation model, open-sourced on Hugging Face on Dec 30, 2025. A 100B+ parameter Mixture-of-Experts architecture (11B active parameters at inference) with a custom Multi-head Latent Attention (MLA) reducing memory usage by up to 83%.

- Achieved state-of-the-art results on Korean language and culture (CLICK), advanced reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [\[Model\]](#) [\[Technical Report\]](#)
- Press: [\[Korea University\]](#) [\[AI Times\]](#) [\[Newstomato\]](#)

Cross-lingual Language Transfer for Large Language Models

2024 – 2025

Lead Researcher (first author)

Published at ACL 2025 (Oral)

- Proposed a cross-lingual optimization method that efficiently transfers English-centric LLMs to new target languages while preserving source-language ability; accepted as an **Oral** presentation at ACL 2025.
- [\[PDF\]](#)

Efficient Reasoning for Large Reasoning Models

2025 – Present

Lead Researcher (first author)

Findings of EMNLP 2026; preprints under review

- Leading a first-author line of work on training-free adaptive thinking budgets (DART) and stable efficiency training via adaptive correct-only rewards (ACOER) for large reasoning models; DART was accepted to **Findings of EMNLP 2026**.
- Co-first author of **ThinkFuse**, a trajectory-aware test-time fusion method for small reasoning models, accepted to **Findings of EMNLP 2026**.
- [\[DART\]](#) [\[ACOER\]](#) [\[Code\]](#)

Awards & Honors

Outstanding Reviewer Award

2026

The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego
Recognized by the ACL program committee for review quality and service to the conference.

Best Paper Award (Top 1)

2025

Annual Conference on Human and Cognitive Language Technology (HCLT 2025)

For “*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*”; **J. Lee** (first author), M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Korea University; Soongsil University; Human-Inspired AI Research.

Minister of Science and ICT Award

2023

ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)

Team M. Kang, **J. Lee**, S. Lee, for “*KoCheckGPT*” — Korean-specialized detection of LLM-generated documents for enterprise security; selected from 79 teams (263 participants) across 28 universities. [\[Press\]](#)

Publications

Underlined = author. * denotes equal contribution. Full profile on [Google Scholar](#).

2026

- “*The Geometry of Hallucination Detection: Why Simple Probes Are Sufficient*”
J. Lee, J. Seo, H. Lim — **EMNLP 2026** (Main)
- “*Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization*”
S. Lee, M. Kim, **J. Lee**, H. Lim — **EMNLP 2026** (Main)
- “*DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models*”
J. Lee, S. Hong, S. Lee, J. Seo, J. Son, S. Eo, C. Park, H. Park, H. Moon, H. Lim — **Findings of EMNLP 2026** [\[arXiv\]](#) [\[Code\]](#)
- “*ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models*”
M. Kang*, **J. Lee***, J. Seo, H. Lim — **Findings of EMNLP 2026** (*equal contribution)

- “Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment”
S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim — **ICLR 2026** (Poster) [\[PDF\]](#)
- “Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation”
C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh — **Knowledge-Based Systems** [\[DOI\]](#)
- “Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards”
J. Lee, S. Lee, S. Hong, M. Kim, C. Park, H. Lim — Preprint, under review [\[arXiv\]](#) [\[Code\]](#)
- “To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG”
J. Lee, C. Park, H. Lim — Preprint, under review [\[arXiv\]](#) [\[Code\]](#)
- “Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations”
D.J. Lee*, **J. Lee***, S. Lee, S. Hong, S. Son, S. Eo, J. Seo, H. Lim — Preprint, under review (*equal contribution) [\[arXiv\]](#) [\[Code\]](#)

2025

- “Cross-lingual Optimization for Language Transfer in Large Language Models”
J. Lee, S. Hong, H. Moon, H.S. Lim — **ACL 2025** (Oral) [\[PDF\]](#)
- “KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification”
J. Lee, M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Annual Conference on Human and Cognitive Language Technology (HCLT 2025) — **Best Paper Award**
- “An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training”
S. Son, C. Park, **J. Lee**, M. Shim, C. Lee, Y. Jang, J. Seo, J. Lim, H. Lim — **Expert Systems with Applications** [\[DOI\]](#)
- “VAETKI Technical Report”
NC AI Consortium — Technical Report [\[PDF\]](#)

2024

- “Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation”
J. Lee, H. Moon, S. Lee, C. Park, S. Eo, H. Ko, J. Seo, S. Lee, H.S. Lim — **Findings of ACL 2024** [\[PDF\]](#)
- “Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0”
C. Park, **J. Lee**, S. Son, K. Park, J. Jang, H. Lim — **Applied Sciences** [\[DOI\]](#)

2023

- “PEEP-Talk: A Situational Dialogue-based Chatbot for English Education”
S. Lee, Y. Jang, C. Park, **J. Lee**, J. Seo, H. Moon, S. Eo, S. Lee, B. Yahya, *et al.* — **ACL 2023** (System Demonstrations) [\[PDF\]](#)
- “A Survey on Evaluation Metrics for Machine Translation”
S. Lee, **J. Lee**, H. Moon, C. Park, J. Seo, S. Eo, S. Koo, H. Lim — **Mathematics** [\[PDF\]](#)
- “Doubts on the Reliability of Parallel Corpus Filtering”
H. Moon, C. Park, S. Koo, **J. Lee**, S. Lee, J. Seo, S. Eo, Y. Jang, H. Kim, H. Lee, *et al.* — **Expert Systems with Applications** [\[DOI\]](#)
- “KoCheckGPT: Korean LLM Written Document Detector”
M. Kang, **J. Lee**, S. Lee, S. Hong, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2023) [\[PDF\]](#)

2022

- “QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation”
S. Eo, C. Park, H. Moon, J. Seo, G. Kim, **J. Lee**, H.S. Lim — **COLING 2022** [\[PDF\]](#)
- “Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing”
H. Moon, C. Park, S. Lee, J. Seo, **J. Lee**, S. Eo, H.S. Lim — **LREC 2022** [\[PDF\]](#)
- “K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set Using Novel Error Type Classification Criteria”
S. Koo, C. Park, J. Seo, S. Lee, H. Moon, **J. Lee**, H. Lim — **IEEE Access** [\[PDF\]](#)
- “Focus on FoCUS: Is FoCUS Focused on Context, Knowledge and Persona?”
S.Y. Lee, **J. Lee**, C. Park, S. Eo, H. Moon, J. Seo, J. Park, H.S. Lim — 1st Workshop on Customized Chat Grounding Persona and Knowledge [\[PDF\]](#)

- “*There is No Rose Without a Thorn: Finding Weaknesses on BlenderBot 2.0*”
J. Lee, M. Shim, S. Son, C. Park, Y. Kim, H. Lim — arXiv preprint [\[arXiv\]](#)
- “*Language Chameleon: Transformation Analysis between Languages Using Cross-lingual Post-training*”
S. Son, C. Park, **J. Lee**, M. Shim, C. Lee, Y. Jang, J. Seo, H. Lim — arXiv preprint [\[arXiv\]](#)
- “*Korean and Multilingual Language Models Study for Cross-lingual Post-training (XPT)*”
S. Son, C. Park, **J. Lee**, M. Shim, C. Lee, K. Park, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)
- “*Policy-based Performance Comparison Study of Real-time Simultaneous Translation*”
J. Lee, H. Moon, C. Park, J. Seo, S. Eo, S. Lee, S. Koo, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)
- “*SaJuTeller: Conditional Generation Deep-Learning based Fortune Telling Model*”
H. Moon, **J. Lee**, J. Seo, S. Eo, C. Park, W. Kim, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)
- “*Korean Language Model Construction and Comparative Analysis with Cross-lingual Post-Training (XPT)*”
S. Son, C. Park, **J. Lee**, M. Shim, S. Lee, J.W. Lee, A. So, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)
- “*DART: Data Augmentation using Retrieval Technique*”
S. Lee, J. Seo, **J. Lee**, M. Kang, H. Moon, C. Park, D. Jung, J. Lee, K. Park, *et al.* — Annual Conference on Human and Language Technology (HCLT 2022) [\[PDF\]](#)

2021

- “*Empirical Study on BlenderBot 2.0's Errors Analysis in Terms of Model, Data and Dialogue*”
J. Lee, S. Son, M. Shim, Y. Kim, C. Park, A. So, J. Park, H. Lim — **Journal of the Korea Convergence Society** [\[PDF\]](#)